

PREDICTING NONRESPONSE IN FUTURE WAVES OF A PROBABILITY-BASED MIXED-MODE PANEL WITH MACHINE LEARNING*

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Nonresponse in panel studies can lead to a substantial loss in data quality owing to its potential to introduce bias and distort survey estimates. Recent work investigates the usage of machine learning to predict nonresponse in advance, such that predicted nonresponse propensities can be used to inform the data collection process. However, predicting nonresponse in panel studies requires accounting for the longitudinal data structure in terms of model building, tuning, and evaluation. This study proposes a longitudinal framework for predicting nonresponse with machine learning and multiple panel waves and illustrates its application. With respect to model building, this approach utilizes information from multiple waves by introducing features that aggregate previous (non)response patterns. Concerning model tuning and evaluation, temporal cross-validation is employed by iterating through pairs of panel waves such that the training and test sets move in time. Implementing this approach with data from a German probability-based mixed-mode panel shows that aggregating information over multiple panel waves can be used to build prediction models with competitive and robust performance over all test waves.

KEYWORDS: Machine learning; Nonresponse; Panel attrition; Prediction; Temporal crossvalidation.

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1. INTRODUCTION

Panel surveys provide a markedly valuable data source for studying social phenomena from a causal perspective by measuring attitudes and (reported) behavior of individuals over time and are therefore used extensively in the social sciences. However, the validity of results drawn from panel data depends on the ability of survey organizations to establish and also to maintain a high-quality panel over time. One of the most severe threats to panel data quality is selective nonresponse of panel members, which can lead to biased survey estimates if response propensities are correlated with the outcome of interest (Groves 2006; Lynn 2009). Moreover, nonresponse may accumulate over multiple panel waves and can therefore be linked to panel attrition, that is, panelists dropping out of the panel altogether (Watson and Wooden 2009). Hence, decreasing sample sizes reduce the efficiency of estimates, limit detailed subgroup analyses and might eventually force survey managers to draw expensive refreshment samples.

Although nonresponse in panel studies has traditionally been tackled by using nonresponse weights (Groves et al. 2009; Rendtel and Harms 2009) or imputation methods (Rubin 1987), recent work studies panel nonresponse from a prediction perspective (Buskirk et al. 2018; Kern et al. 2019). In this context, machine-learning methods (ML; for an overview, see Hastie et al. 2009) are employed, following an increasing number of studies that utilize advantages in the field of machine/statistical learning to tackle social science research problems (e.g. Mullainathan and Spiess 2017; Molina and Garip 2019). Hence, nonresponse prediction aims at identifying nonrespondents in advance, shifting the focus from post- to pre-correction of panel nonresponse. The prediction models are typically sought to inform the data collection process, akin to the idea of adaptive designs (Groves and Heeringa 2006; Lynn 2017), for example, by targeting likely nonrespondents in new panel waves with higher incentives. Against this background, prediction accuracy in future panel waves becomes the center of attention.

Previous work indicates that ML may be used to accurately predict attrition and nonparticipation for selected hold-out waves or samples, particularly in the context of (probability-based) online panels with frequent data collection (e.g. Klausch 2017; Lugtig and Blom 2018). However, there is little guidance on how to efficiently train, tune, and evaluate nonresponse prediction models in a longitudinal context. This study proposes a framework for building and evaluating nonresponse models that particularly focus on incorporating information from multiple panel waves. With respect to model building, this approach utilizes features that aggregate previous (non)response patterns, for example, by counting the total number of nonparticipations up to a given wave. Concerning model tuning and evaluation, temporal cross-validation (*rolling forecasting origin* technique, Hyndman and Athanasopoulos 2018) is employed by iterating through pairs of panel waves. This allows for tracking the performance of prediction models in multiple test sets and selecting the model (i.e., the combination of model type, hyperparameter setting, and feature groups) that performs

best over time. Hence, the evaluation approach mimics the application of the prediction models in the field and enables performance comparisons under deployment conditions. That is, completely new sets of panel waves are used for performance evaluation and historical information up to a given wave is used for model training. The proposed framework draws on the model building and evaluation schemes that are implemented in the general purpose risk modeling and prediction toolkit *triage* (Crockett et al. 2018).

This study illustrates the longitudinal perspective on nonresponse prediction with data from a German probability-based mixed-mode online and mail access panel (GESIS Panel; Bosnjak et al. 2017). We consider penalized logistic regression, decision trees, random forests, extremely randomized trees, and extreme gradient boosting for building the prediction models. We construct multiple feature groups by aggregating time-variant characteristics over different time frames (considering only the last wave, the last three waves or all previous waves) to study the effect of including historical information on prediction performance. Altogether, this results in 4,000 prediction models that are built and evaluated over 20 training and test waves.

We discuss the prediction models with respect to their ability to identify likely nonrespondents in advance, in line with the higher-level goal of developing a prediction-based targeted design. To this end, we study the prerequisites of an effective targeted design for reducing nonresponse in a probability-based mixed-mode panel: How accurately can different models differentiate between future respondents and nonrespondents? Which types of ML methods are most effective? Which (longitudinal) feature groups achieve optimal prediction performance? It should be noted that given this research perspective, this study focuses solely on the model building and evaluation step that precedes an actual implementation of ML-based targeting.

The remainder of this article is structured as follows. Section 2 briefly reviews previous approaches for modeling and predicting nonresponse in panel studies. Section 3 lays out the proposed framework and, in parallel, its application, that is the data (section 3.1), the feature generation process (section 3.2), the temporal crossvalidation strategy (section 3.3), and the model types that are used in the empirical example (section 3.4). The results are presented in section 4, which includes identifying and selecting optimal model settings over time (section 4.1) and a more detailed evaluation of the selected models in the last test wave (section 4.2). We close by discussing the advantages and limitations of our longitudinal prediction approach in section 5.

2. ANALYZING NONRESPONSE AND ATTRITION IN PANEL STUDIES

Nonresponse and attrition in panel studies have been analyzed with a multitude of statistical methods in previous research, depending on the objective of the

respective modeling task. On a higher level, these objectives may be classified as *explanation*, which typically entails the usage of parametric methods, and *prediction*, which turns the focus toward supervised learning techniques.

A common approach for studying factors that affect nonresponse is to model contact and cooperation propensities with logistic regressions on a wave-by-wave basis, typically by focusing on predictors from the respective previous wave (see e.g. Lepkowski and Couper 2002; Behr et al. 2005; Lipps 2007; Siegers et al. 2019). As this can lead to many analyses, the probability of responding in a given wave may be analyzed longitudinally on the person-wave level, which allows the inclusion of time as a covariate (Behr et al. 2005; Uhrig 2008; Lipps 2009). A related approach is to particularly study the time dependence of attrition with survival analysis (e.g. Richter et al. 2014; Struminskaya and Bosnjak 2015). Further extensions include modeling the response process as the occurrence of sequential events (establishing contact with the sample member and obtaining co-operation given successful contact) by using bivariate probit models (Nicoletti and Peracchi 2005; Watson and Wooden 2009) or analyzing the count of complete interviews per year with multilevel negative binomial regression (De Leeuw et al. 2017).

Another set of studies focuses on modeling different types of response patterns, for example by creating a multiclass outcome which differentiates between attrition at different points in time (i.e., in earlier versus later panel waves, Durrant and Goldstein 2010). Although this approach covers monotonic attrition patterns, further work separates attriting from returning respondents in multinomial regressions to study the factors that affect temporary nonresponse (Burkam and Lee 1998; Voorpostel 2010). This perspective is extended by Lugtig (2014) who categorizes respondents based on the similarity of their response patterns using latent class analysis.

In this context, it is worth noting that studies that model panel nonresponse with parametric regression methods also commonly include information about response patterns in previous waves as predictors. This includes adding panel experience, temporary drop-out in previous waves, or the absolute number of previously completed surveys to the set of explanatory variables (Roßmann and Gummer 2016; Siegers et al. 2019). Related work studies the effects of changes and events between waves on panel attrition (Voorpostel and Lipps 2011; Trappmann et al. 2015). A thorough inclusion of historic information is implemented by Kocar (2019) who constructs longitudinal predictors by computing average survey outcome rates (for participation, noncontact, and refusal) prior to a wave, the number of consecutive survey outcomes prior to a wave, and changes in response status over all previous waves.

Although the outlined studies typically focus on studying the factors that cause nonresponse and attrition in panel surveys, recent work approaches nonresponse from a prediction perspective. Initial work in this field primarily investigates and compares the usage of different ML methods with different prediction objectives. Kern et al. (2019) predict refusals in the German Socio-

Economic Panel Study with one pair of panel waves and report favorable prediction performance of tree-based ensemble methods in comparison with logistic regression. [Klausch \(2017\)](#) focuses on short- and long-term attrition in the Longitudinal Internet Studies for the Social Sciences (LISS) panel and demonstrates remarkable prediction performance independent of the prediction method (comparing lasso, random forest, and BART models, data in wide format). Also using the LISS panel, [Mulder and Kieruj \(2018\)](#) report favorable performance of random forests compared to multilevel logistic regression (data in long format). In contrast, [Lugtig and Blom \(2018\)](#) find little differences in the performance of logistic regression and random forests when predicting attrition in the German Internet Panel (GIP) on a wave-by-wave basis. Further examples include [Liu and Wang \(2018\)](#) who predict reparticipation in the Surveys of Consumers, [McKay \(2019\)](#) who predicts nonresponse in Understanding Society, [Würbach and Zinn \(2019\)](#) who predict participation status in the NEPS Newborn Cohort, and [Bach et al. \(2020\)](#) who estimate response propensities for the LISS panel, SOFT, and EPBG survey.

It should be noted, however, that the aforementioned studies use different methods and metrics to evaluate prediction performance which limits the comparability of results. More importantly, previous studies that use ML for predicting nonresponse do not make full use of the longitudinal data structure in terms of model building and evaluation. Against this background, we propose to employ feature generation by aggregating over time in connection with temporal crossvalidation as a general framework for building and evaluating nonresponse prediction models with panel data.

3. METHODS

3.1 Data

In this study, data from the GESIS Panel ([Bosnjak et al. 2017](#)) are used to exemplify the longitudinal prediction approach. The GESIS Panel is a probability-based mixed-mode access panel that is based on a random sample from German population registers, started in 2013 and is conducted on a bi-monthly basis. In the initial recruitment sample, the target population consisted of the German-speaking population between 18 and 70 years. The initial sample of the GESIS Panel was drawn from the residents' registration offices with a two-stage selection procedure (and the selection stages being communities and individuals). The selection of the communities was realized in proportion to their size. All persons in this initial sample have been sent a questionnaire every 2 months since 2013, unless they left the panel. There is no further selection of panelists in or out of individual panel waves. The GESIS Panel comprises two participation modes, that is, an online as well as a mail mode. An incentive of 5 euros is sent out by mail along with the invitation letter to all

(i.e., online and offline) panelists. After the invitation, two e-mail reminders are sent to the online panelists if they have not yet participated in the respective survey.

The GESIS Panel is an access panel, which means that it is open to the academic research community for data collection. Researchers from universities and noncommercial research institutes can submit a proposal containing questions/items for about 5 minutes of survey time. The studies (currently >40) can be cross-sectional or longitudinal. Some questions, for example on demography, are part of a core module which is usually repeated annually. The topics of the core studies and their inclusion frequencies are outlined in the online [Supplementary table A.1](#). For a complete list of submitted studies and their contents, we refer to the GESIS Panel study descriptions ([GESIS Panel 2020](#)).

In this study, we use data that start with the first complete wave of the GESIS Panel (i.e., the first survey that is fielded with the complete sample after recruitment), wave ba (February 2014), and ends with wave ed (August 2017), which results in a total of 22 panel waves ([GESIS 2018](#), the waves are labeled based on the following naming scheme: [year: a = 2013, b = 2014, ...] [waves: a, ... f = 1 ... 6]). In addition, we use information from the recruitment interview which was conducted in 2013.

As in other panel studies, the GESIS Panel sample is subject to attrition, which can be caused by various mechanisms. On the one hand, participants drop out of the panel if, for example, they leave the panel owing to illness or death. Panelists can also unsubscribe from the panel themselves if they no longer wish to participate (voluntary attrition). In addition, participants will be excluded if they have not participated in a wave at least three times in a row (involuntary attrition). Panelists cannot re-join the GESIS Panel if they dropped out at one point in time.

It should be noted that to partly compensate for the effects of panel attrition in the GESIS Panel, a refreshment sample is recruited every 2 years. In the following analysis, however, we will solely focus on the first cohort of the GESIS Panel, which provides data over the longest time span. Specifically, our analysis includes active panel members of the first cohort for each wave, that is panelists who permanently drop out in a given wave are excluded from our analysis in the following waves. As a result, we start with a sample size of $n_{ba} = 4,888$ in wave ba (February 2014) and end with $n_{ed} = 3,286$ in wave ed (August 2017).

3.2 Variables

The main objective of this study is to predict nonresponse in the GESIS Panel on a wave-by-wave basis. More precisely, we use a binary variable that indicates participation (0) or nonparticipation (1) as our outcome for each panel wave. We define participation as a complete or partially complete interview

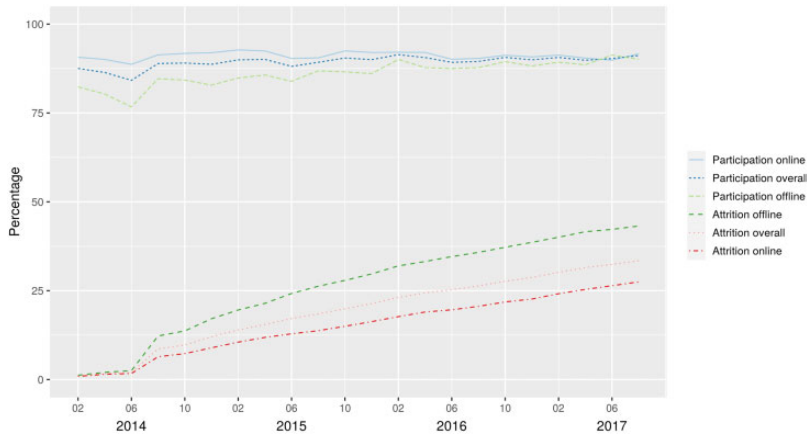


Figure 1. Cumulative Percentage of Attrition and Participation Rates in the GESIS Panel.

with sufficient information (AAPOR categories 11 and 12; [The American Association for Public Opinion Research 2016](#)) and consider refusal, non-contact and other outcomes as nonparticipation (following AAPOR RR6; [The American Association for Public Opinion Research 2016](#)). In wave ed (August 2017), the AAPOR categories with the highest frequencies that combine into nonparticipation of the then active panel members were “319: Nothing ever returned” (7.39 percent), “212: Break-off” (0.67 percent), and “211221: Logged on to survey, did not complete any items” (0.46 percent). The proportion of noncontact was 0.24 percent (AAPOR category 3311), overall resulting in a participation rate of 91.15 percent ([Schaurer and Schmidt 2018](#)).

To provide some context for the following analysis, [figure 1](#) shows attrition and participation rates in the GESIS Panel by mode for our study period. The three lines in the lower part of the plot display the cumulative percentage of attrition (offline, overall, and online) relative to the active panel population before wave ba (February 2014) was fielded (i.e. all panelists who completed the GESIS Panel welcome survey). We compute the attrition rate as follows:

$$\text{Attrition rate}_{\text{wave } i} = \frac{\text{Overall active panel}_{\text{welcome survey}} - \text{Overall active panel}_{\text{wave } i}}{\text{Overall active panel}_{\text{welcome survey}}}$$

It can be seen that offliners are more likely to leave the panel than online participants. Furthermore, there is a strong increase in attrition between wave bc (June 2014) and wave bd (August 2014). This is due to the fact that at this point persons are excluded from the panel owing to three consecutive nonresponses for the first time. In the upper part of the plot, it becomes apparent that the overall participation rate (complete or partially complete interviews relative

to all panelists invited for a given wave, see the definition of our outcome variable above) initially fluctuates quite strongly and then stabilizes at around 90 percent from 2015 onward. The participation rate of online users is almost always higher than the participation rate of offliners, whereas the two rates converge toward each other over time. With participation rates hovering around 90 percent, we observe class imbalance in our outcome variables (i.e. a higher proportion of participation (0) than nonparticipation (1)) that needs to be accounted for in the model evaluation process.

Nonparticipation in panel waves has been linked to a variety of factors that operate on different stages of the fielding process (locating the sample member, contacting the sample member, obtaining cooperation; Lepkowski and Couper 2002; Watson and Wooden 2009). As nonparticipation in the GESIS Panel can be considered to be predominantly driven by non-co-operation and our longitudinal setup requires us to focus on predictors that are available for all panel waves, we use *respondent characteristics* and previous *interview experience* as main explanatory concepts. Hence, we follow the previous studies that identified co-operation in previous waves and enjoyment and perceived difficulty of previous interviews as important drivers of nonresponse and attrition (e.g. Hill and Willis 2001; Olsen 2005; Frankel and Hillygus 2014).

The predictor variables (in the ML literature these variables are also called “features”) are summarized in table 1 (for question wordings and response categories, see the online Supplementary tables A.2–A.5). With respect to respondent characteristics, we focus on socio-demographic variables and general personal traits. These variables were derived from the recruitment interview of the GESIS Panel and are treated as *time-invariant* in the following analysis. With respect to (previous) experience of interview, we consider cooperation-related variables from the recruitment interview (e.g. interviewer assessments on the willingness to participate in the GESIS Panel) as well as response status, survey evaluation (e.g., whether the questionnaire was considered as too long) and general survey participation (e.g., whether the respondent answered the survey in one piece) from each panel wave. We further include binary missing indicators that flag missing values for each predictor variable (i.e., 1 if information for a predictor is missing in a given wave, 0 otherwise). It should be noted that this allows us to include panelists who temporarily do not participate in a given (prior) wave, but who are still part of the active panel population, in our models.

The wave-specific variables are, by definition, *time-variant* and therefore require some form of pre-processing scheme to be efficiently used for predicting nonresponse in a given wave. We create features from time-variant variables by aggregating over time, that is summing the occurrence of a given category in previous waves. This results in, for example, counting the number of complete interviews in the last three waves (for each wave). When constructing longitudinal features, we consider different time horizons to compare the predictive power of short- (last wave, Block II), medium- (last three waves, Block

Table 1. Predictor Variables

Concept	Variables	Source
Respondent/socio-demographic characteristics	Age, sex, migration background, education, marital status, household size, employment status, job type, personal income, household income, house type, house condition, social status, life satisfaction, general trust	Recruitment interview
Survey co-operation	Survey experience, willingness to respond (in recruitment interview), willingness to participate (in recruitment interview), willingness to participate (in panel), probability of participation (in panel), provided telephone number, provided e-mail address	Recruitment interview
Response status	Response status (complete interview, partial interview, nonparticipation)	Panel waves (ba-ed)
Survey evaluation	Questionnaire: Interesting, diverse, important for science, long, difficult, too personal, overall assessment	Panel waves (ba-ed)
Survey participation	Mode, participation interrupted, participation location	Panel waves (ba-ed)

III), and long-term (all previous waves, Block IV) response histories. With this setup, we structure our complete set of predictors in four blocks:

- Block I: Time-invariant
 - Respondent/socio-demographic characteristics from recruitment interview
 - Survey cooperation in recruitment interview
- Block II: Time-variant (last wave)
 - Response status, survey evaluation, and participation in last wave
- Block III: Time-variant (aggregated, last three waves)
 - Response status, survey evaluation, and participation over the last three waves
- Block IV: Time-variant (aggregated, all previous waves)
 - Response status, survey evaluation, and participation over all previous waves

These feature blocks are built for each wave of the GESIS Panel, akin to generating an outcome variable for each panel wave. It should be noted that as the last block aggregates over all previous waves at a given point in time, the amount of information that is summarized in this block increases as we move toward more recent panel waves. When building the prediction models, we

allow models to use all feature blocks simultaneously and each block separately, which results in five feature groups (i.e. Block I, II, III, IV, and all). This allows us to study the effect of aggregating over multiple panel waves with different time horizons on prediction performance.

3.3 Temporal Crossvalidation

A key requirement for evaluating and comparing nonresponse prediction models with panel data is that the evaluation method closely aligns with the intended usage of these models in the field. Commonly, the goal is to predict nonresponse in a new panel wave based on information from previous waves. This task corresponds to the objective of forecasting with time series data for which various evaluation methods have been proposed (see Tashman 2000; West 2006; Bergmeir and Benitez 2012). Common techniques include fixed-origin evaluation, where the time point which separates the fit and test period—the origin—is fixed, and rolling-origin evaluation, where the origin moves forward in time (Hyndman and Athanasopoulos 2018). Temporal crossvalidation resembles the rolling origin technique by structuring training and test sets by time, such that—in a panel data setting—completely new sets of panel waves are used for performance evaluation and historical information is used for training. Repeating this process allows for tracking the performance of a given prediction model over time, enabling model selection based on performance trends. In this study, the temporal crossvalidation results are used to narrow down the set of trained models and to eventually identify the combination of model type, hyperparameter setting, and feature groups that consistently results in the best performance.

Figure 2 shows the temporal cross/validation setup of this study. The first training set draws on features from the recruitment interview (Z) and from the first complete wave of the GESIS Panel (X_1^{train} , February 2014) and builds the outcome (i.e., nonparticipation) based on wave two (y_1^{train} , April 2014). The first test set uses features from the recruitment interview (Z) and from the first two GESIS Panel waves (X_1^{test} , February and April 2014) and builds the outcome based on wave three (y_1^{test} , June 2014). It should be noted that for all X -blocks, the same number of variables is created each time, whereas the information that is included and aggregated in those variables are time dependent. Also, it should be noted that the outlined procedure deliberately introduces a dependence between the training and the test sets by using similar sets of (active) panel participants in all folds—in our panel data setting, the “new” test data for model deployment include *known* participants for which we want to infer nonresponse risks based on past (non-)response trajectories. In this context, the temporal crossvalidation setup ensures that prediction models are trained with feature information that are given up to a specific point in time and evaluated with respect to their performance of predicting the outcome in a

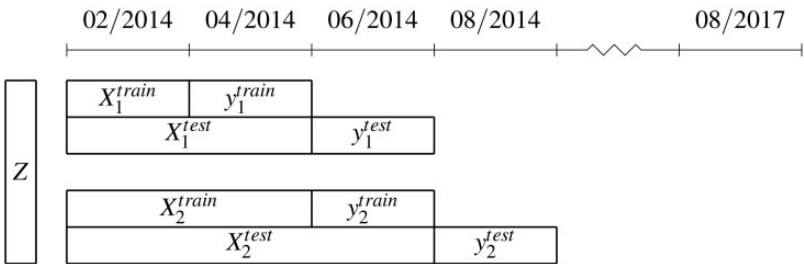


Figure 2. Temporal Crossvalidation with the GESIS Panel.

new panel wave. This process is repeated up to wave ed (August 2017), that is additional pairs of training and test sets are created by moving forward in time which results in 20 training and 20 test sets in total (note that testing starts in wave three).

Temporal crossvalidation allows for mimicking a variety of specific use cases of predictive modeling with longitudinal data (<https://dssg.github.io/triage/>, last accessed February 23, 2021). Although this application focuses on predicting one wave ahead, multiple extensions such as aggregating outcomes over multiple waves or allowing a time gap between the features and the outcome can be implemented. We further chose to retrain models for each new panel wave that becomes available, whereas other model update frequencies could be investigated as well. Finally, in our setup, each model is trained by considering only one row of data or one outcome event per panelist, respectively. An alternative approach would be to incorporate multiple outcome events per panelist, which would result in more observations for each training set, but also in a more complicated data structure (long format with events nested in panelists).

3.4 Model types

We use the outlined temporal crossvalidation schema to compare the performance of a number of prediction models that are built with different ML methods. Following previous studies on nonresponse prediction, we focus on tree-based methods that are typically well-suited for prediction tasks with many features and potentially complex nonadditive and/or nonlinear relationships between the predictors and the outcome of interest (Klausch 2017; Kern et al. 2019). In this context, we include single decision trees (CART) as a rather simple and interpretable approach as well as prominent (random forests) and more recent (extremely randomized trees, XGBoost) ensemble methods that have been shown to perform well in a variety of settings (Caruana and Niculescu-Mizil 2005; Fernández-Delgado et al. 2014; Chen and Guestrin 2016). We

further consider penalized logistic regression to also include parametric “benchmark” models. In total, five model types are studied:

- Penalized Logistic Regression
 - Logistic regression with a (lasso/ridge) penalty on the vector of regression coefficients (Tibshirani 1996)
- Decision Trees
 - Recursive partitioning technique, repeatedly splits predictor space into homogeneous subregions (Breiman et al. 1984)
- Random Forest
 - Tree-based ensemble method, grows decorrelated decision trees based on bootstrap samples (Breiman 2001)
- Extremely Randomized Trees (ExtraTrees)
 - Tree-based ensemble method, grows trees based on randomly sampled predictors and cut points at each node with the full training data (Geurts et al. 2006)
- Extreme Gradient Boosting (XGBoost)
 - Sum-of-trees method, builds a sequence of trees akin to optimization via gradient descent with each tree attempting to improve over the result of its predecessor (Chen and Guestrin 2016)

Each prediction method is associated with a set of hyperparameters that has to be tuned to achieve optimal performance. This includes, for example, determining the optimal size of a decision tree or the number of trees that are used in an XGBoost model. We consider exhaustive grid search with the hyperparameter settings that are outlined in table 2, that is, individual models will be built for each combination of the considered hyperparameter settings. It should be noted that we include nearly unpenalized logistic regression models in our model set by including a high value for the inverse of the regularization strength (C) in the tuning grid.

In the longitudinal context, the present tuning strategy results in building one (for example) random forest for each tuning parameter setting and training set and then evaluating the performance of random forests with this setting over time. As we also consider different feature groups (section 3.2), a prediction model for a given model type and training set consists of a combination of hyperparameter settings and feature groups that are included. All in all, this results in 200 models (i.e. 5 feature groups $\times$ 40 tuning parameter settings) that are built for each training set, that is, 4,000 (200×20) unique models in total.

The computational infrastructure for implementing the analysis is provided by the Python (3.6.4) package triage (2.2.0, Crockett et al. 2018) and draws on PostgreSQL (9.5.17) for data management and scikit-learn (0.19.1, Pedregosa et al. 2011) and xgboost (0.90, Chen and Guestrin 2016) for model building. Data preparations, model selection and postmodeling are conducted with R (3.4.4, R Core Team 2018). Our R and Python code is available at <https://osf>.

Table 2. Tuning Grids

Model type	Hyperparameter	Values	<i>n_{settings}</i>
Logistic regression	Penalty	11, 12	8
	C	0.05, 0.1, 1, 1000	
Decision trees	max_depth	3, 5, 10	6
	max_features	null, sqrt	
Random forest	max_features	sqrt, log2	4
	min_samples_leaf	1, 10	
	n_estimators	500	
Extra trees	max_features	sqrt, log2	4
	min_samples_leaf	1, 10	
	n_estimators	500	
XGBoost	max_depth	3, 5, 10	18
	n_estimators	250, 500, 1,000	
	learning_rate	0.1, 0.05	
	Subsample	0.8	

NOTE.— Scikit-learn default settings are used for parameters that are not listed.

io/pbuzs/?view_only=265488386bb342c7a7829abffd93b2a7 (last accessed February 23, 2021). The Scientific Use File of the GESIS Panel is publicly available at the GESIS data archive (GESIS 2018).

4. RESULTS

We structure the results section by first investigating the prediction performance of the trained models over the full set of test sets with respect to different performance metrics (section 4.1). Besides summarizing overall performance levels, given all available features, section 4.1 discusses selecting optimal combinations of feature groups and hyperparameter settings for each model type. In a production setting, this aligns with condensing the full set of models to a smaller set of candidates, and eventually, to just one final model that would be implemented in the field. We then evaluate and compare the selected best models with respect to prediction performance, similarities of the predicted lists, and feature importances in the last test wave (section 4.2).

4.1 All Waves

Owing to the large number of models that were built in each training set, we start by presenting prediction performances only for models that use all feature blocks, that is models that were allowed to draw on all available information.

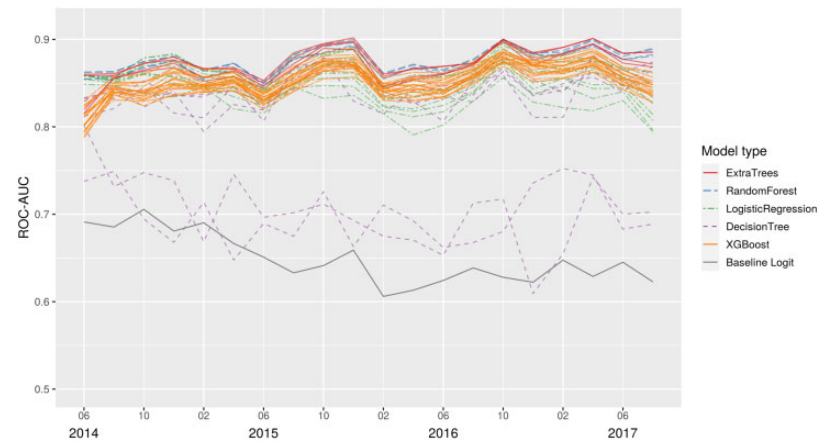


Figure 3. ROC-AUCs for All Waves and Models with All Feature Blocks. *Note:* Each line represents a unique combination of model type and hyperparameter setting. Lines with the same color represent models of the same type, but with different hyperparameter settings.

Figure 3 shows the development of ROC-AUCs (area under the receiver operating characteristic curve) of these models over all test sets, grouped by model type. It should be noted that ROC-AUC is a more robust measure than, for example, accuracy for prediction problems with class imbalance (Kuhn and Johnson 2013) and varies between [0,1], with 0.5 representing a noninformative model and 1 representing a model that achieves perfect separability. First, it can be seen that the highest ROC-AUCs that can be reached by any model type range roughly between 0.85 and 0.9, indicating strong prediction performance among the top models. Second, the performance curves do not follow a clear (up- or downward) trend over time although selected test waves are associated with lower average ROC-AUCs (e.g. wave cc, June 2015). Third, the top performance curves include random forest, extra trees and penalized logistic regression models, followed by XGBoost and decision tree models (on average). Finally, almost all models achieve considerably higher performance levels when compared to unregularized “baseline” logit models that use only time-invariant characteristics as features (feature block I, gray line), indicating a strong benefit of including features that aggregate over multiple panel waves (the online [Supplementary table A.6](#) lists logit coefficients of the first (trained 04/14) and last (trained 06/17) baseline logistic regression). However, there are two notable exceptions. The two lines that hover around ROC-AUC scores of about 0.7 on average represent decision tree models with a maximum depth of 10, indicating that those trees tend to overfit, that is were grown too deep for the prediction task at hand.

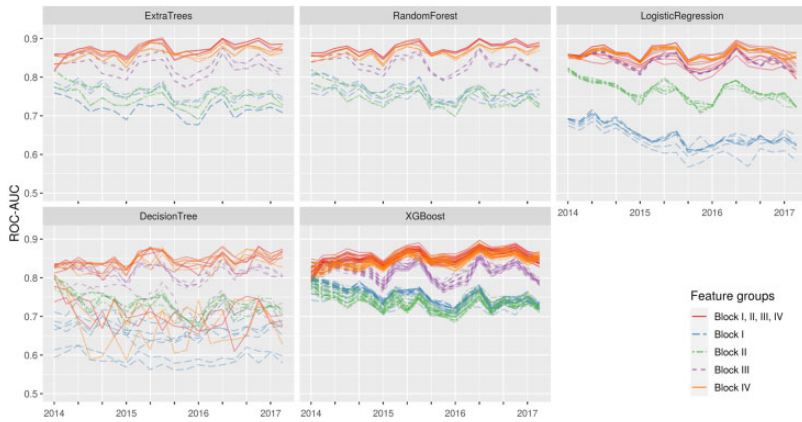


Figure 4. ROC-AUCs by Model Type and Feature Group. *Note:* Each line in each subfigure represents a unique combination of hyperparameter setting and feature groups used. Lines with the same color represent models which use the same feature groups, but different hyperparameter settings.

The effect of in- or excluding different feature blocks on model performance is shown in figure 4. It can be seen that a large share of variance in ROC-AUCs for a given model type and test set is induced by different feature group settings. Excluding features that aggregate over multiple previous waves (feature block III, IV) typically leads to performance curves with comparably low average ROC-AUCs (blue and green lines). In most cases, the best performance levels are achieved by combining all feature blocks, whereas the strongest contribution to ROC-AUC seems to come from feature block IV (aggregating over all previous waves). Furthermore, figure 4 shows that performance differences between feature blocks III and IV increase after a couple of test waves, as the amount of information that is aggregated into block IV increases over time.

The reported performance trends can be used to select the optimal hyperparameter and feature group combination for each model type. For this purpose, we first compute the average ROC-AUC for all models up to the second to last test set, enabling performance comparisons in the last test set under deployment conditions in later steps. On this basis, we follow a simple “best average” criterion by selecting the respective hyperparameter and feature group setup that is associated with the highest mean ROC-AUC for each model type. It should be noted that other selection rules could be used by, for example, accounting for the variance of a given performance metric or by giving more weight to more recent test sets. In our results, we observe relatively stable ROC-AUCs over time (with the exception of some decision tree models), especially when conditioning on the feature groups used (see figure 4 again). The performance results of the selected models and their

hyperparameter settings are summarized in the online [Supplementary table A.7](#). We will discuss these models in more detail in section 4.2.

The online [Supplementary figure A.1](#) compares models that use all feature blocks with respect to precision at top 5 percent and top 10 percent as alternative evaluation metrics. In this context, panelists with predicted nonresponse probabilities that are among the highest 5 percent (10 percent) of all scores are predicted as being nonrespondents and then evaluated against the true classes (positive predicted value; true positives/predicted positives). That is, we focus on the proportion of true nonrespondents among all predicted nonrespondents, based on two classification thresholds (the 95th and 90th percentile of the predicted nonresponse probabilities, respectively). We include an example calculation of precision at top 10 percent and recall at top 10 percent for the best selected random forest model and the last test set in the online [Supplementary table A.8](#). These performance metrics align with the intervention perspective in which panelists with the highest predicted risk scores may be targeted in an adaptive design, given two hypothetical scenarios in which the panel management has resources to target 5 percent or 10 percent of all panelists with, for example, higher incentives. In this context, low precision would indicate a misallocation of resources such that mostly those panelists who would have participated anyway would receive higher incentives. It should be noted that the thresholds were selected based on hypothetical resources rather than based on the empirical distribution of nonrespondents—if our aim was to maximize the proportion of detected nonrespondents (i.e., recall), we would focus on less strict thresholds which, however, comes at the cost of lower precision. For both measures, random forests, extra tree models and penalized logistic regressions are among the best performing models while also being associated with relatively low variance in performance for a given test set. The temporal cross-validation results show a decreasing trend in the best precision values over time, indicating that prediction-based targeting using the two classification thresholds at hand may be based on an increasing number of false positives in future waves. This could potentially be counterbalanced by using more restrictive classification thresholds, which, however, likely reduces recall (see the discussion of recall below). However, for the last test set and best models, about 45–50 percent of the panelists that are predicted as being nonrespondents based on the 10 percent cutoff are truly nonrespondents (the online [Supplementary table A.7](#)), which are still markedly higher numbers than the baseline performance (precision equals to the proportion of nonrespondents; $290/3,286 \times 100 = 8.83$ percent in the last wave).

Complementing the online [Supplementary figure A.1](#), the online [Supplementary figure A.2](#) shows recall at top 5 percent and top 10 percent for models that use all feature blocks. This evaluation focuses on the proportion of nonrespondents that are detected among all nonrespondents in a given wave when applying the same classification thresholds as before (sensitivity; true positives/positives in test data). That is, we are interested in the proportion of

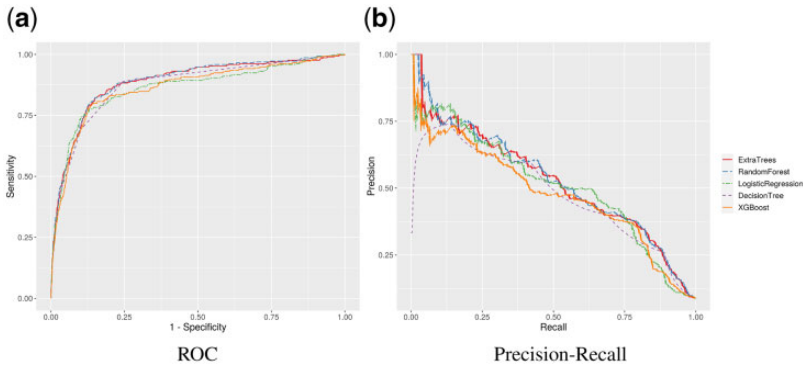


Figure 5. Performance Curves of Best Models For Most Recent Test Wave. (a) ROC (b) Precision–Recall

nonrespondents that we are able to detect in a scenario in which we intervene only on a small percentage of panelists owing to limited resources. It becomes apparent that similar patterns as with previous metrics can be observed although the recall curves do not follow a clear (up- or downward) trend over time. Recall percentages between 51 and 57 percent for the last test set and best models based on the top 10 percent cutoff illustrate that focusing on high risk observations with a restrictive threshold leads to relatively precise predictions (see above), but misses a large share of nonrespondents that are present in the data (the online [Supplementary table A.7](#)).

4.2 Most Recent Wave

After summarizing overall performance trends and selecting optimal settings for each model type, we focus on more detailed analyses of the respective best models in the last test set. Hence, we follow a potential implementation scenario of an ML-based intervention in which a final best model would be used to target likely nonrespondents in the most recent wave of the GESIS Panel (for example) with higher incentives after model selection based on temporal crossvalidation.

Figure 5 shows ROC and precision–recall curves, which plot sensitivity versus one minus specificity and precision versus recall over the full range of applicable classification thresholds for the best models in the most recent test set. The selected candidate models achieve high prediction performance with ROC–AUCs between 0.86 (penalized logistic regression, XGBoost) and 0.89 (random forest, extra trees), indicating that after model tuning and feature group selection all model types perform comparably well (see also the online [Supplementary table A.7](#)). It should be noted that this also holds true for the best, but relatively simple, decision tree model. A similar result is given by the

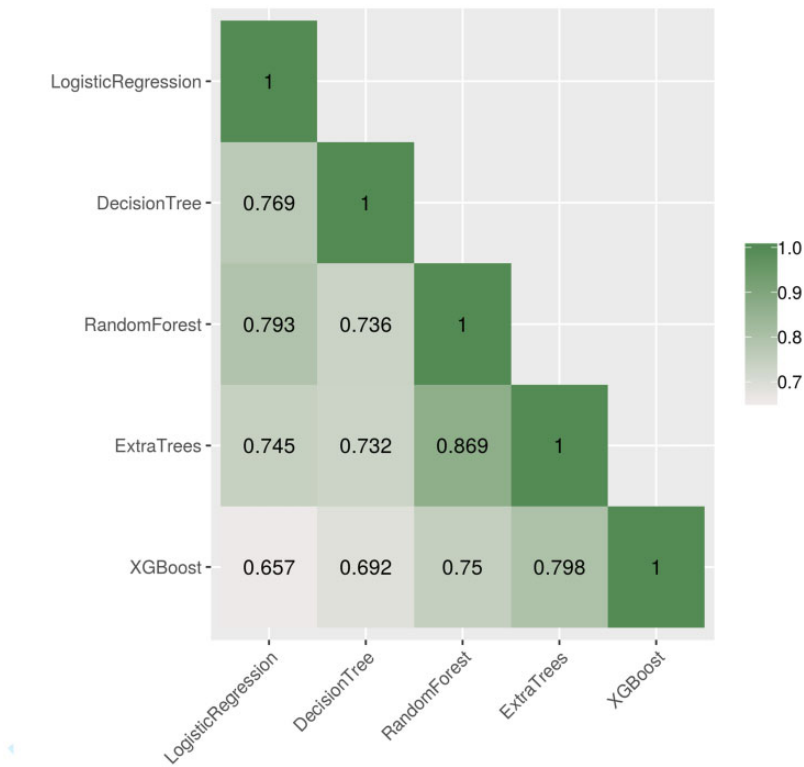


Figure 6. Jaccard Similarities Between Predicted Lists (Top 10 Percent).

precision–recall curves, where all models considerably improve over the precision baseline of a random classifier (which equals to 8.83 percent for any classification threshold).

The selected models can also be compared with respect to the resulting lists of panel members with particularly high nonresponse risks. This allows us to evaluate whether different optimal models result in similar sets of panelists who might be targeted in a prediction-based intervention. We generate lists of panelists-at-risk for each model by classifying individuals with predicted risk scores that are among the highest 10 percent of all scores as likely nonrespondents. We then compute Jaccard similarities (Tan et al. 2018) between these lists,

$$J(L_a, L_b) = \frac{|L_a \cap L_b|}{|L_a| + |L_b| - |L_a \cap L_b|}$$

with $|L_a|$, $|L_b|$ being the total number of predicted nonrespondents in list a , b and $|L_a \cap L_b|$ being the number of panelists that are predicted as

nonrespondents in both lists. J varies between $[0,1]$, with higher values indicating more similar lists. The computed Jaccard similarities are shown in [figure 6](#). It can be seen that the best prediction models do not only result in similar performance, but also produce similar lists of likely nonrespondents with large overlap. Maybe not surprisingly, the strongest agreement occurs between random forests and extra trees which produce lists that are almost interchangeable.

Finally, the selected models can be compared with respect to the importance of different types of features in the context of model building. The online [Supplementary figure A.3](#) displays mean importance scores for each feature block and feature concept combination ([table 1](#)), including a “na” category that summarizes the importance of indicator variables that flag missingness. The underlying importance scores aim to measure the “effect size” for the predictor variables and correspond to absolute logit coefficients for penalized logistic regression, improvement in accuracy for XGBoost, and Gini importance for the remaining model types. All importance scores have been scaled to have a maximum value of 100. On the one hand, the online [Supplementary figure A.3](#) indicates that the tree ensembles draw on a more diverse set of features compared to the logistic regression and decision tree model, as shown by the fewer spots of zero mean importances. On the other hand, the highest mean importances for the tree ensembles are achieved by feature sets that include aggregated missing indicators (i.e., sums of missings over different time spans). As these variables carry the information that a panelist did not participate in a given wave (but was still part of the active panel population and therefore included in our models), response status in previous waves appears to be by far the most important factor when building the present prediction models.

In summary, given the average ROC–AUC performance, the relative robustness over time and the (small) edge in prediction performance in the most recent wave, both the random forest and the extra trees models which use the full feature set (and `max_features = log2`, `min_samples_leaf = 10`, see the online [Supplementary table A.7](#)) are likely the best candidates that could be chosen for implementing a prediction-based targeted design in the GESIS Panel.

5. DISCUSSION

This study presented and applied a longitudinal framework for building and evaluating prediction models to predict nonresponse in panel studies based on multiple panel waves and ML. This approach particularly focuses on providing effective prediction models that can be used to target likely nonrespondents in future panel waves in the context of an adaptive design. Using data from the GESIS Panel, it has been shown that generating features by aggregating over multiple panel waves can be used to build models with notable prediction performance. Furthermore, temporal crossvalidation was used to select and

evaluate prediction models in a data setting that mimics the prospective usage of these models in the field. The GESIS Panel results indicated that after parameter tuning and feature group selection all model types (penalized logistic regression, decision trees, random forests, extremely randomized trees, and extreme gradient boosting) performed comparably well for identifying prospective nonrespondents in the present application, with ROC–AUCs between 0.86 and 0.89, precision at top 10 percent between 45 and 50 percent and recall at top 10 percent between 51 and 57 percent for the best models in the last test wave. It was also shown that the prediction models particularly benefited from including response status information from multiple previous waves as features, and that performance differences were largely driven by in- or excluding different feature groups rather than by different tuning parameter settings (especially for the ensemble models). That is, in terms of prediction performance, the set of predictor variables used generally mattered more than the model type and the respective hyperparameter setting.

The presented analyses can be extended in various ways. First, we used a simple binary outcome that only distinguished between participation and nonparticipation in a given panel wave. A refined analysis would partition nonparticipation in subclasses that align with different potential interventions (e.g., for noncontact and refusal) or could model individual trajectories by focusing on transitions to first nonparticipation and to final drop-out, again enabling different treatments. This can be paired with predicting over various time horizons, as different interventions might need different time frames for implementation. Second, the set of predictor variables could be extended by, for example, including more substantive variables on topics that are included in multiple waves, building features from online paradata for the web respondents or by accounting for changes in socio-demographic characteristics over time. Third, the presented feature aggregation approach could be compared with an alternative strategy that directly includes every measurement of a given variable as predictors. It should be noted, however, that longitudinal feature aggregation has a similar benefit of allowing models to give stronger weights to more recent features when feature blocks that aggregate over different time spans are included. Fourth, we considered rather small tuning grids with a limited number of try-out values for model training and better performance of, for example, the XGBoost models may be achieved with more sophisticated tuning settings. In addition, model tuning and selection solely focused on prediction performance, whereas the key objective of a prediction-based intervention would be to increase response rates *and* to decrease bias owing to group-specific response propensities. An extended application could therefore consider model selection based on prediction performance and bias reduction, for example by including dissimilarity indices (Roßmann and Gummer 2016) or R-indicators (Schouten et al. 2012) as selection criteria. The GESIS Panel also offers additional means of performance evaluation such as building and cross-validating prediction models across different cohorts of the panel (e.g.

evaluating the performance of the selected best models of this study with data from the first refreshment sample).

Furthermore, it is important to note that although the presented prediction framework can generally be implemented with data from any panel study, this article focused on its application with one specific data source. Although the GESIS Panel shares several design aspects with other studies (e.g., LISS, GIP, ELIPSS, see [Blom et al. 2016](#)), additional work is needed to learn about the generalizability of the presented results. These applications would hence allow one to gather insights about which design features drive performance differences across panels, that is for which type of panel prediction-based targeting might be most successful. As an example, as response status information from multiple previous waves were among the most important features, it is doubtful that similarly high performance levels could be achieved in the context of more traditional panel studies with longer time periods between panel waves (e.g., one wave per year as opposed to every other month). In this case, however, the presented approach could be (more) fruitful to predict other types of outcomes, such as noncontact.

6. CONCLUSION

Finally, we want to stress that building a highly accurate and carefully selected prediction model is only the first step in the process of developing a prediction-based targeted design. This study exclusively focused on model building and evaluation without utilizing the obtained predictions to ultimately target likely nonrespondents. Mitigating nonresponse in practice requires both accurate prediction models and effective interventions. The adaptive design literature discusses a wide range of design features that may be altered based on nonresponse predictions, with mixed results regarding their effectiveness ([Lynn 2017](#)). Among those options, differential incentives might be viewed as the most promising avenue for increasing response rates among groups with low predicted participation propensities ([Zagorsky and Rhoton 2008](#); [Zhang 2010](#)). However, as panel studies might already offer some form of baseline incentives to all panelists as part of their panel maintenance strategy, additional rewards that are targeted might be less effective ([Mercer et al. 2015](#)). A further complication is that respondents with low response propensities may be worse reporters, that is persuading reluctant respondents to participate in a panel wave might increase measurement error ([Bach et al. 2020](#)). A prediction-based intervention would therefore need careful evaluation and long-term monitoring with respect to different aspects of data quality to study the potential benefits and risks of this approach.

Supplementary Materials

Supplementary materials are available online at academic.oup.com/jssam.

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